VU Medical Zone

BT 501 Midterm M.C.Q

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1. Process or practice - modifies or harnesses any living organism or system to be useful to any human purpose. **Technology**

2. In Bioremediation _____ are used as biosensor. **Bacteria and Fishes**

3. Penicillin is discovered by Scottish scientist **Sir Alexander Fleming** in 1928.

4. Which one of the earliest discovered antibiotics? **Penicillin**

5. Penicillin is only most effective drug against **Gram +ve** bacteria.

6. Type 1 diabetes mellitus- insulin-producing cells become dysfunctional -less amount of insulin - elevation of the **blood sugar level.**

7. _____ produced from debilitating or killed forms of the microorganisms. **Vaccine**

8. **1796** Edward Jenner found a milkmaid infected with smallpox.

9. Which one of the following prevent from an upcoming infection? **Prophylactic**

10. _____ proteins by the immune system to target specific foreign object (antigen). **Antibody**

11._____ an antibody by a single clone of cells identical antibody molecules. **Monoclonal Antibody**

12. _____ gives cells structure on which they need to grow, without those cells are free floating, cannot connect with each other, communicate or form tissue. **Scaffold**

13. **Medicine researchers** face many difficulties in constructing suitable scaffolds.

14. A cell that has the ability to continuously divide and differentiate (develop) into various other kind(s) of cells/tissues. **Stem cells**

15. _____ can be used to cure chronic blood-related disorders such as sickle cell disease, Thalasemia and leukemia. **Hematopoietic stem cells (HSC), Mesenchymal stem cells (MSC)**

16. **Amniotic** stem cells can differentiate in cells of adipogenic, osteogenic, myogenic, endothelial, hepatic and also **neuronal lines**.

17. For autologous transplants physicians now usually collect stem cells from the **peripheral blood** rather than the **marrow**

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18. _____ is introduction of new adult stem cells into damaged tissue in order to treat disease or injury. **Stem cell therapy**

19. Stem cells show great promise for regenerative **medicine**

20. _____ are pluripotent stem cells derived from the inner cell mass of a blastocyst, an early-stage pre-implantation embryo. **Embryonic stem cells (ES cells)**

21. Human embryos reach the **blastocyst stage 4–5 days** post fertilization, at which time they consist of **50–150 cells**.

22. _____ results in destruction of the blastocyst, which raises ethical issues, including whether or not embryos at the pre-implantation stage**.** **Inner cell mass (ICM)**

23. ___ generates the outer layer of the embryo and produces the surface layer (epidermis) of the **skin** and forms the **nerves**. **Ectoderm**

24. **Endoderm** becomes the innermost layer of the embryo and produces the **digestive tube** and its associated organs (including the lungs).

25. **Mesoderm** becomes sandwiched between the ectoderm and endoderm and generates the blood, heart, kidney, gonads, bones, and connective tissues.

26. Cell therapy in which the stem cells in a donor's marrow are used to replace the blood cells of the victims of **leukemia**.

27. Cell therapy is also being used in experiments to **graft new skin cells** to treat serious burn victims, and to **grow new corneas** for the sight-impaired.

28. What Diseases can be cured by Stem Cell Therapies? **Alzheimer's disease, Parkinson's disease, Spinal cord injury, Heart disease, Severe burns and Diabetes.**

29. International Human Genome Sequencing was founded in **1998.**

30. Human genome ~**3.2 Gb**.

31. Only **28%** DNA is transcribed into RNA.

32. Only **1.1%-1.4%** of genome actually encodes protein (=5% of transcribed RNA).

33. Called co-suppression because suppressed expression of both **endogenous gene** and **transgene**.

34. RNAi components also involved in: **Transposon silencing, Viral defense and Gene regulation**

35. Discovery of DNA structure Watson & Crick in **1953.**

36. **Isolation of DNA ligase in 1967**

37. **Isolation of REase in 1970**

38. **Paul Berg generated rDNA technology in 1972**

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39. **Cohen & Boyer in 1973 produced first plasmid vector**

40. ____ is artificial insulin production, drug delivery to target sites. **Pharmacology**

41. The microchip stores a unique identification number from **10 to 15** digits long.

42. How many types f transponder? **Two**

43. A **biochip** can store and update financial, medical, demographic data, basically everything about a person.

44. **Gene therapy** is a technique for correcting **defective genes** responsible for disease development.

45. An abnormal gene could be swapped for a normal gene through **homologous recombination.**

46. **4 Types** of vectors currently used in gene therapy.

47. ___ creation of an artificial lipid sphere with an aqueous core. **Liposome**

48. Chemical linking chemically linking the **DNA** to a molecule that will bind to special cell receptors.

49. The first gene therapy clinical trial began in **1990**.

50. Scientist at the National Institutes of Health (Bethesda, MD) have successfully treated metastatic melanoma in two patients using **killer T cells** genetically retargeted to attack the cancer cells.

51. RNA interference or gene silencing may be a new way to treat **Huntington's.**

52. Anytime a foreign object is introduced into human tissues, the immune system is designed to attack the invader. **Immune response.**

53. Multigene disorder disorders that arise from **mutations** in a **single gene** are the best candidates for gene therapy.

54. How many types of somatic gene therapy? **Two**

55. **Liposomes** are concentric bilayered vesicles in which an aqueous.

56. Liposomes were discovered about **40 years** ago by **Alec Bangham.**

57. Liposomes are micro particulate **lipoidal vesicles** which are under extensive investigation as drug carriers for improving the delivery of therapeutic agents.

58. Liposomes have been widely investigated for the treatment of both anterior and posterior segment **eye disorders**.

59. **Bio-Nanotechnology** is a division of nanotechnology which uses biological starting materials.

60. **Nanotherapeutics** is an application of bionanotechnology in which the diseases can be treated by the use of various techniques at **molecular level**.

61. ____ are the vectors which are widely used for the genes delivery into the target cells. **Harmless Virus**

62. Stem cells are the **second** type of vectors that are used by the scientists.

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63. **Immunotherapy** is defined as the treating the disease by inducing, enhancing, or suppressing an immune response.

64. By 2015, the global nanotechnology market is expected to achieve **$1 trillion**.

65. A thread-like structure of nucleic acids and protein found in the nucleus of most living cells, carrying genetic information in the form of genes. **Chromosome**

66. Which one of the following risk factors for chromosomal abnormalities? **Radiation, Smoking and Alcohol consumption**

67. Structural chromosome rearrangements are changes in the physical structure of chromosomes that may result in-: **birth defects, mental retardation, increased risk for infertility and pregnancy loss**.

68. Abnormalities in which a portion of chromatin from a single chromosome is lost are called **deletions**.

69. Duplications are unbalanced rearrangements that result in partial **trisomy**.

70. _____ in which the long arms of two acrocentric chromosomes are joined with loss of the two short arms. **Robertsonian translocations**

71. **Inversions** are formed by two breaks in the same chromosome with exchange of the two ends.

72. **Paracentric- chromosome** is formed when both breaks occur in the same arm and, therefore, the centromere is not included in the inverted segment.

73. Chromosome non-disjunction is defined as the unbalanced segregation of chromosomes leading to **aneuploidy**.

74. ____ sleep disorder characterized by pauses in breathing or instances of shallow or infrequent breathing during sleep. **Sleep apnea**

75. _____ decline in mental ability severe enough to interfere with daily life, memory loss. **Dementia**

76. ____ the process by which most genes on one of the two X chromosomes in females are silenced epigenetically and fail to produce any product. **X chromosome inactivation**

77. Klinefelter syndrome is relatively common among infertile males **(about 3%)** or males with oligospermia or azoospermia **(5% to 10%).**

78. Klinefelter syndrome patients are **tall and thin** and have relatively long legs.

79. A **single gene disease** occurs due to a change in a gene that leads to loss or disturbed function of that gene.

80. Examples of common gene disorders are the **hemoglobinopathies** and **cystic fibrosis**.

81. Heterozygous mothers have a **50%** chance to transmit the gene to each son or daughter.

82. An affected with autosomal dominant condition usually has **one** affected parent.

83. Boys are affected by **X-linked recessive** condition when they get the abnormal allele from their carrier mother.

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